

CHAPTER TWO

INTELLIGENT AGENTS

CoSC 412

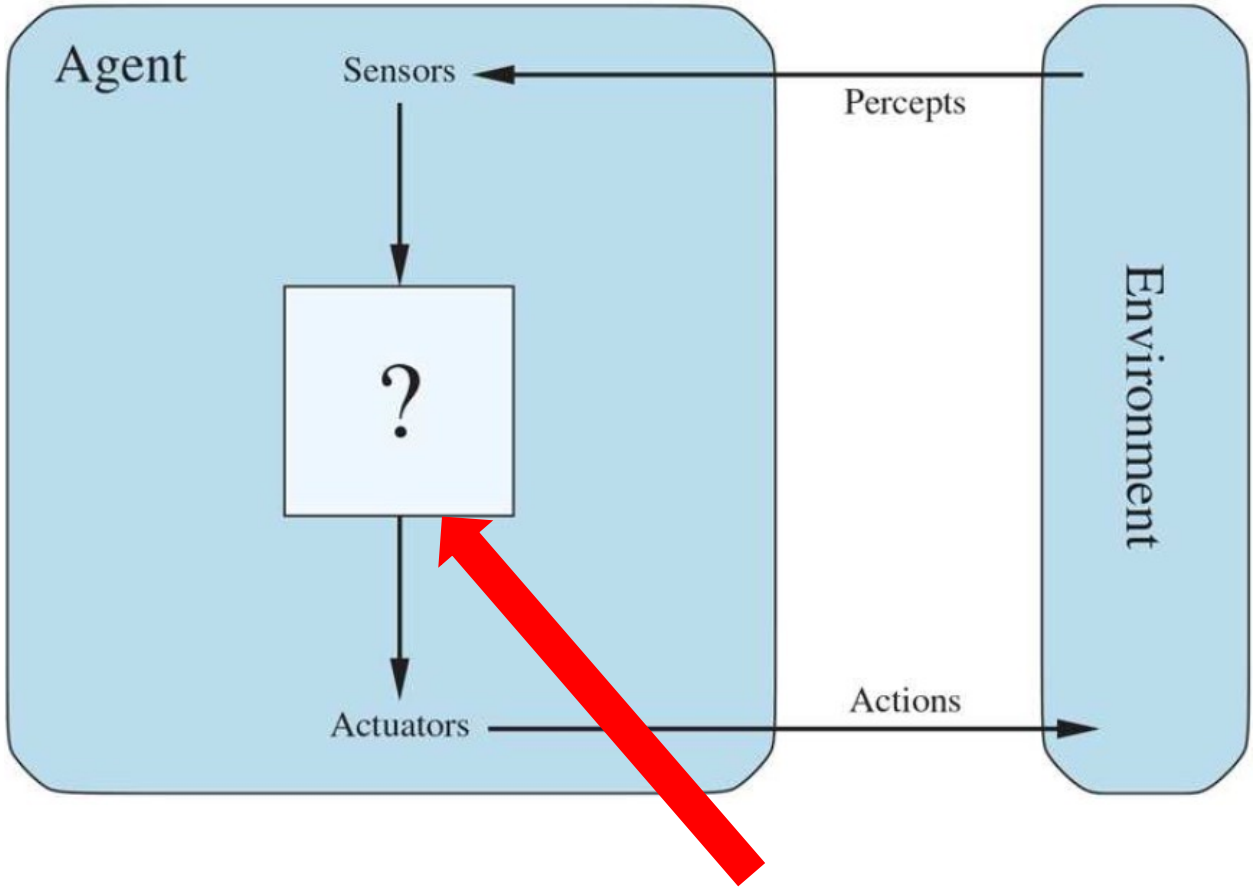
Ashenafi W.

Intelligent Agents

- **Agent** is one that:
  - *Perceives* via sensors (Observes its environment) – Camera, any other input
  - Takes action in its environment via its actuators

- Environment is that which is outside of the agent but affects the agent and is also affected by it
  - Limited part of the entire universe (for obvious purposes)

- Examples:
  - Humans can be thought of as agents
    - Inputs – eyes, ears
    - Actuators/effectors – Hands, legs
  - Robotic agent
    - Inputs/sensors – cameras, infrared sensors,
    - Actuators – motors
  - Software Agent
    - Inputs – from keyboards, mouse, packets, etc.
    - Outputs – manipulate files, display info, etc.



What do you think should come here?

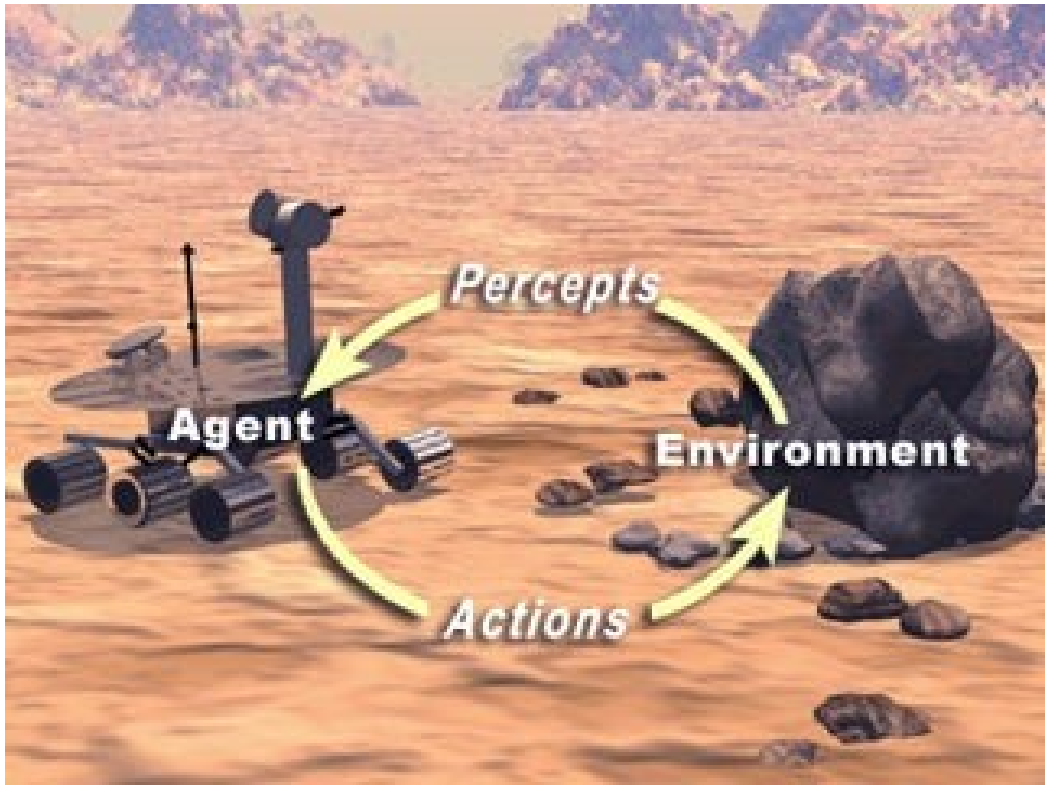
Intelligent Agents – Some terminologies

- Percepts:
  - The content an agent's sensors are perceiving

- Percept Sequence:
  - Complete history of everything the agent has ever perceived (up to the present)

- In a give situation/scenario an agent makes a decision based on its
  - ***built in knowledge*** (that which was original)
  - ***Precept sequence*** (learned over time)

- **Agent function** which maps any given percept sequence to an action describes an agent's behavior
  - Externally it's a table of actions (a list)
  - Internal to the agent the process is a program



Example of Agent Types

Agent type	Percepts (input)	Actions	Goals (output)	Environment
Medical diagnosis system	Symptoms, findings, patient's answers	Questions, tests, treatments	Healthy patients, minimize costs	Patient, hospital
Satellite image analysis system	Pixels of varying intensity, color	Print a categorization of scene	Correct categorization	Images from orbiting satellite
Part-picking robot	Pixels of varying intensity	Pick up parts and sort into bins	Place parts in correct bins	Conveyor belts with parts
Refinery controller	Temperature, pressure readings	Open, close valves; adjust temperature	Maximize purity, yield, safety	Refinery
Interactive English tutor	Typed words	Print exercises, suggestions, corrections	Maximize student's score on test	Set of students

How should an agent act?

- **Rational agent**
  - One that does the right thing.
  - if the system achieves the objective assigned

- Which actions are correct? (Rationalistic Action approach)
  - The actions that cause the agent to be most successful
- Therefore, success of an agent should be measured in someway

- What is rational at any given time depends on **four things**:
  - The *performance measure* defining the criterion of success
  - The agent's *prior knowledge* of the environment
  - The actions that the agent can perform
  - The agent's percept sequence up to now

How should an agent act? - Continued

- For each possible percept sequence, a rational agent should select an action expected to maximize its performance measure,
  - Given the evidence provided by the percept sequence and whatever built-in knowledge the agent has
  - For example, during exam objective is to maximize marks, based on the questions on the paper & your knowledge



Learning

- Does a rational agent depend on only current percept?
  - No, the past percept sequence should also be used. this is called **<u>learning</u>**
  - After experiencing an episode (event), the agent should adjust its behaviors to perform better for the same job next time.



Autonomy

- If an agent just relies on the **prior knowledge** of its designer rather than its **own percepts** then the agent lacks *autonomy*
- *A rational agent should be **autonomous -** it should learn what it can to compensate for partial or incorrect prior knowledge.*
- E.g. a clock

  No input (percepts)

  Run only but its own algorithm (prior knowledge)

  No learning, no experience, etc.

Software Agents

- Sometimes, the environment may not be the real world
  - E.g. flight simulator, video games, Internet
  - They are all artificial but very complex environments
  - Those agents working in these environments are called Software agent (softbots) Because all parts of the agent are software

Task Environments

- Sometimes, the environment may not be the real world
  - E.g. flight simulator, video games, Internet
  - They are all artificial but very complex environments
  - Those agents working in these environments are called Software agent (softbots) Because all parts of the agent are software
- Specifying the task environment, PEAS description as fully as possible
  - **P**erformance measure
  - **E**nvironment
  - **A**ctuators
  - **S**ensors

- In designing an agent, the first step must always be to specify the task environment as fully as possible.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

PEAS description of the task environment for an automated taxi.

Task Environments – Continued

- ***Performance measure***
  - How can we judge the automated driver?
  - Which factors are considered?
    - getting to the correct destination
    - minimizing fuel consumption
    - minimizing the trip time and/or cost
    - minimizing the violations of traffic laws
    - maximizing the safety and comfort, etc.

- ***Actuators***
  - Control over the accelerator, steering, gear shifting and braking
  - A display to communicate with the customers

- ***Environment***
  - A taxi must deal with a variety of roads
  - Traffic lights, other vehicles, pedestrians, stray animals, road works, police cars, etc.
  - Interact with the customer

- ***Sensor***
  - camera,

Properties of task environments - Continued

- ***Fully observable vs. Partially observable***
  - ***Fully observable***
    - If an agent's sensors give it access to the complete state of the environment at each point in time then the environment is **effectively and fully observable**
  - **Partially observable**
    - If an agent's sensors doesn't give it access to the complete state of the environment.
    - Because of noisy and inaccurate sensors or parts of the state are simply missing from the sensor data.
    - **E.g.** A local dirt sensor of the cleaner cannot tell Whether other squares are clean or not.

- ***Deterministic vs. stochastic***
  - Next state of the environment Completely determined by the current state and the actions executed by the agent, then the environment is ***deterministic***,
  - If the environment is not completely determinable/erratic, then **it is Stochastic**.
  - **E.g.** Cleaner and taxi driver are: Stochastic because of some unobservable aspects: noise or unknown



Properties of task environments - Continued

- ***Episodic vs. sequential***
  - ***An episode*** = agent's single pair of perception & action
  - The quality of the agent's action does not depend on other episodes
  - Every episode is independent of each other
  - Episodic environment is simpler
  - The agent does not need to think ahead
  - **Sequential**
  - Current action may affect all future decisions
  - E.g. Taxi driving and chess.

- ***Static vs. dynamic***
  - ***A dynamic environment*** is always changing over time
    - E.g., the number of people in the street
  - ***Static environment***
    - E.g. the destination
  - ***Semi dynamic***
    - environment is not changed over time
    - but the agent's performance score does

Properties of task environments - Continued

- ***Discrete vs. continuous***
  - If there are a limited number of distinct states, clearly defined percepts and actions, the environment is **discrete**
    - E.g. Chess game
  - **Continuous environment**
    - E.g. Taxi driving



- ***Known vs. unknown***
  - *In known environment*, the outcomes for all actions are given.
    - Example: solitaire card games.
  - *If the environment is unknown*, the agent will have to learn how it works in order to make good decisions.
    - Example: new video game.

Properties of task environments - Examples

Task Environment	Observable	Deterministic	Episodic	Static	Discrete	Agents
Crossword puzzle	Fully	Deterministic	Sequential	Static	Discrete	Single
Chess with a clock	Fully	Strategic	Sequential	Semi	Discrete	Multi
Poker	Partially	Strategic	Sequential	Static	Discrete	Multi
Backgammon	Fully	Stochastic	Sequential	Static	Discrete	Multi
Taxi driving	Partially	Stochastic	Sequential	Dynamic	Continuous	Multi
Medical diagnosis	Partially	Stochastic	Sequential	Dynamic	Continuous	Single
Image-analysis	Fully	Deterministic	Episodic	Semi	Continuous	Single
Part-picking robot	Partially	Stochastic	Episodic	Dynamic	Continuous	Single
Refinery controller	Partially	Stochastic	Sequential	Dynamic	Continuous	Single
Interactive English tutor	Partially	Stochastic	Sequential	Dynamic	Discrete	Multi

Examples of task environments and their characteristics.

Structure of AI Agents

- An intelligent agent is composed of
  - Architecture – underlying machinery the AI agent executes/performs its function
  - Agent Function – maps a percept/environmental input into a an action
    - f:P* → A
- And then there's the *agent program,* which implements the agent function by running on the agent architecture to produce f
  - Input for Agent Program - Only the current percept
  - Input for Agent Function - The entire percept sequence
    - The agent must remember all of them
  - Implement the agent program as a *look up table* (**agent function**)

- An agent based on a prespecified lookup table.
  - It keeps track of the percept sequence and just looks up the best action.

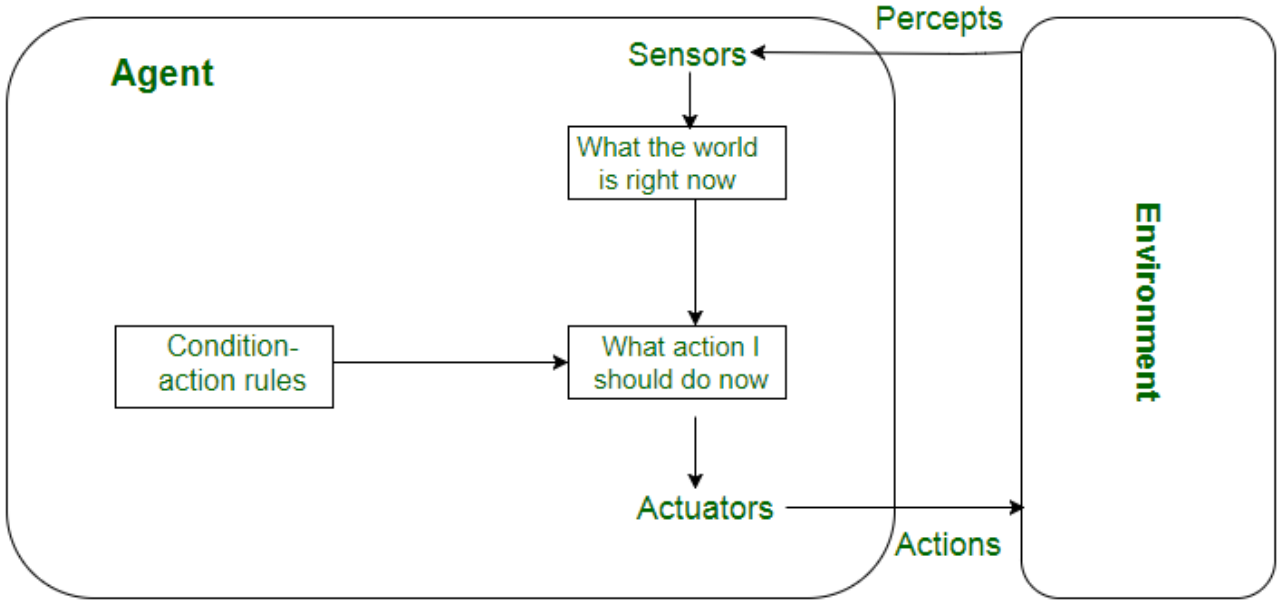
function TABLE-DRIVEN-AGENT(*percept*) **returns** an action
  persistent: *percepts*, a sequence, initially empty
    table, a table of actions, indexed by percept sequences, initially fully specified

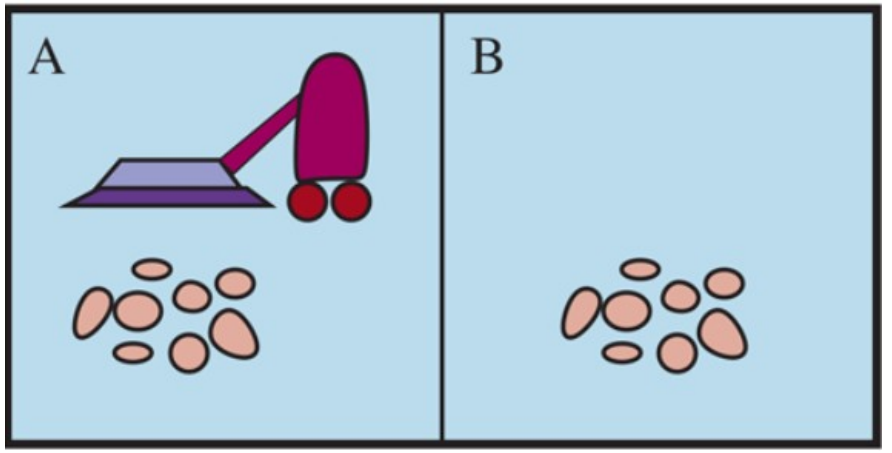
  append *percept* to the end of *percepts*
  action ← LOOKUP(*percepts*, *table*)
  return *action*

The TABLE-DRIVEN-AGENT program is invoked for each new percept and returns an action each time. It retains the complete percept sequence in memory.

Types of agent programs

- Based on degree of perceived intelligence (from our perspective) and capability
  - Simple Reflex Agents: act only upon the current percept (don't consider percept seq.)
    - Here agent function is based on the condition-action rule (maps a state to an action)
    - If the condition is true, then the action is taken, else not
    - Agent function works only in a fully observable environment
    - E.g. – Vacuum agent (has dirt sensor and location sensor)
  - How should the vacuum behave if it's location sensor fails? (Doesn't know where it's head is)
  - Some sort of guessing is needed –
    - If clean then randomly choose left/right





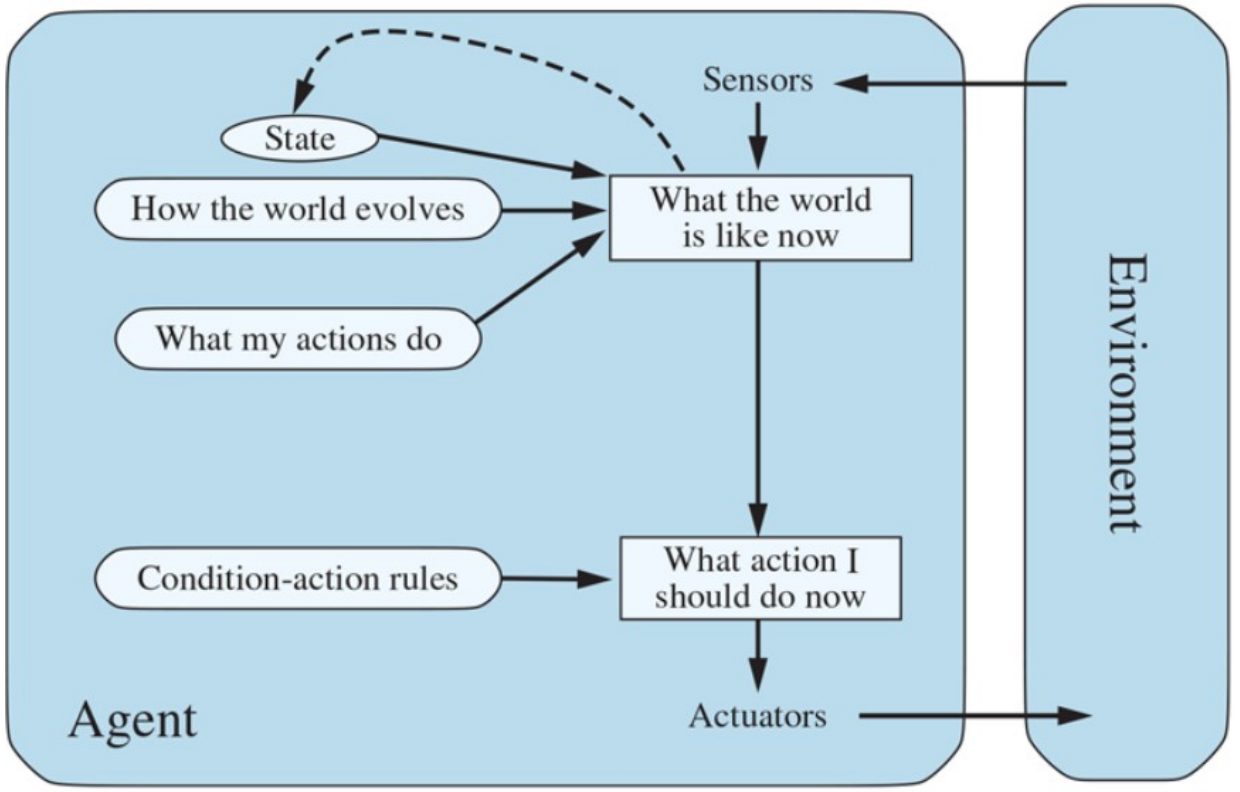
function REFLEX-VACUUM-AGENT([*location*,*status*]) **returns** an action

if *status* = *Dirty* **then return** *Suck*
else if *location* = *A* **then return** *Right*
else if *location* = *B* **then return** *Left*

The agent program for a simple reflex agent in the two-location vacuum environment. This program implements the agent function

Types of agent programs

- Based on degree of perceived intelligence (from our perspective) and capability
- The most effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now
  - **Model-based reflex agents**: have
    - ***Internal state:*** depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state
    - ***Transition Model***: Knowledge about how the world works
      - Effects of the agent's actions
      - How the world evolves independently of the agent
    - ***Sensor Model:*** perception of the world
  - *Model based agent uses transition and state model to keep track of its environment – given the fidelity of its sensors*



Types of agent programs

- Based on degree of perceived intelligence (fror our perspective) and capability
  - *Model based Reflex Agent – Continued*
    - ***Transition Model***:

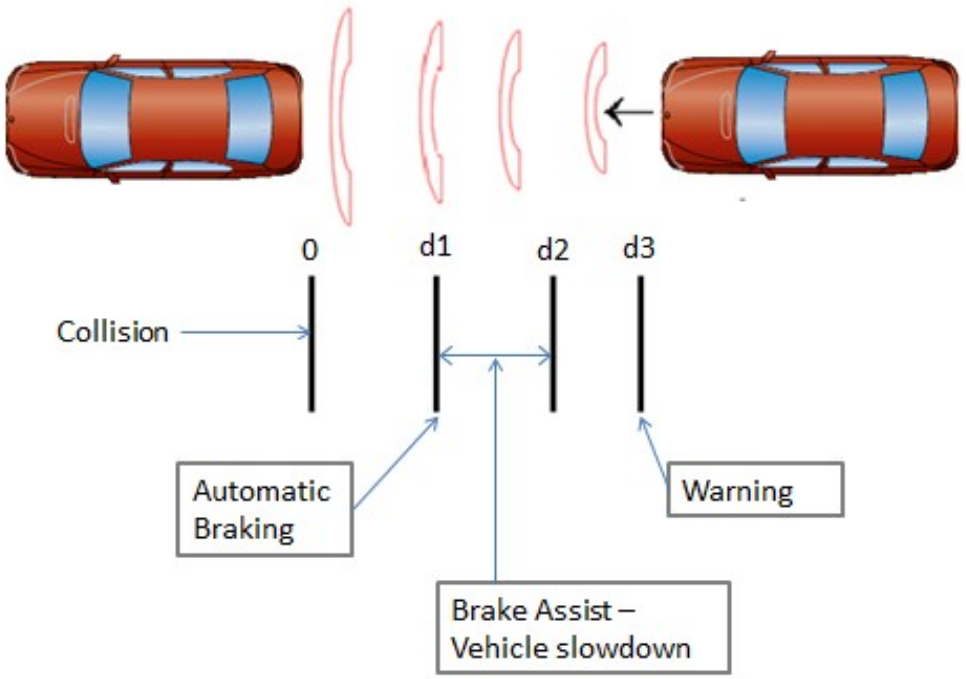
    For example, when the agent turns the steerin wheel clockwise, the car turns to the right, When it's raining the car's cameras can get we
    - ***Sensor Model***:

    For example, when the car in front initiates braking, one or more illuminated red regions appear in the forward-facing camera image, and, when the camera gets wet, droplet-shaped objects appear in the image partially obscuring the road
    - ***Internal State:***

    For example: internal state is not too extensive —just the previous frame from the camera, allowing the agent to detect when two red lights at the edge of the vehicle go on or off simultaneously

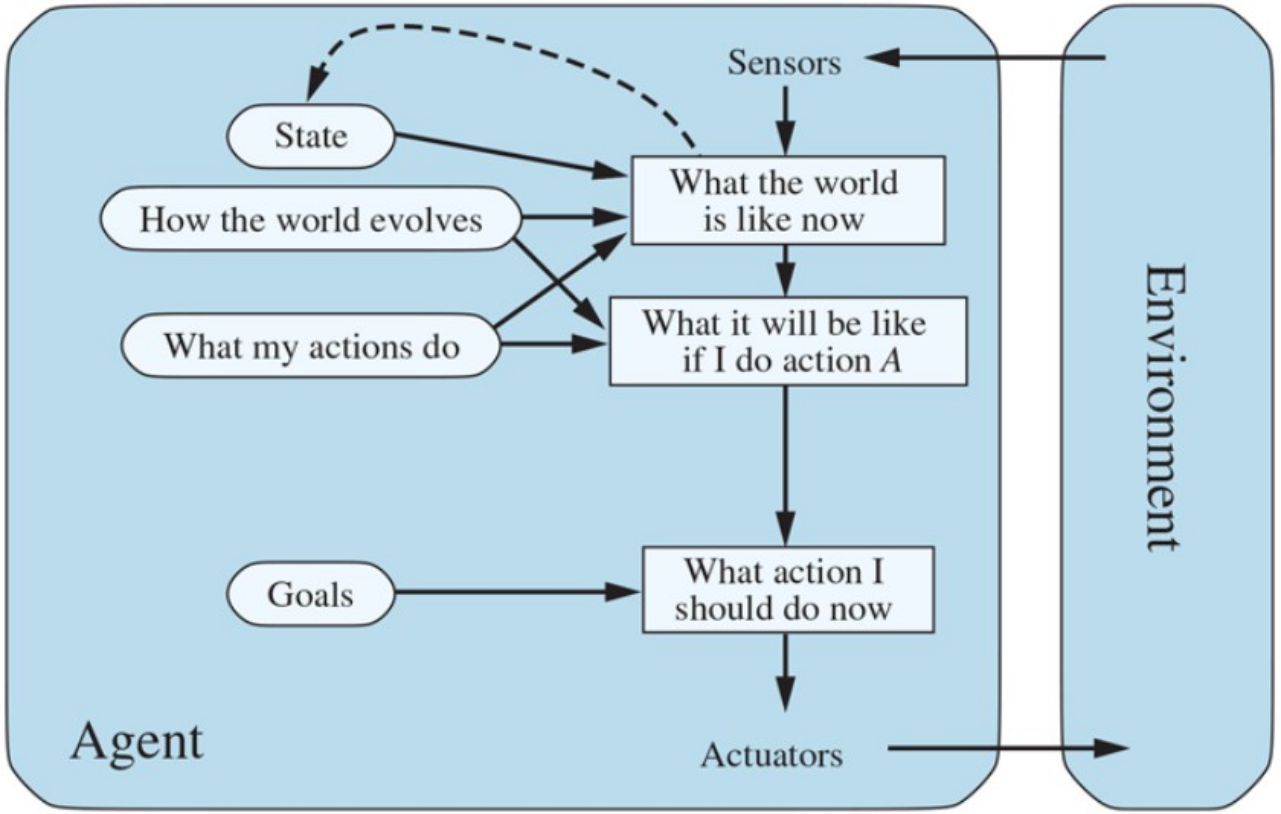
function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action
persistent: *state*, the agent's current conception of the world state
transition_model, a description of how the next state depends on the current state and action
sensor_model, a description of how the current world state is reflected in the agent's percepts
rules, a set of condition–action rules
action, the most recent action, initially none

state ← UPDATE-STATE(*state*, *action*, *percept*, *transition_model*, *sensor_model*)
rule ← RULE-MATCH(*state*, *rules*)
action ← *rule*.ACTION
return *action*



Types of agent programs

- Based on degree of perceived intelligence (from our perspective) and capability
  - ***Goal-based Agent**:* as well as a current state description (Model-based), the agent needs some sort of **goal** information that describes situations that are desirable
  - Combine model and goal to achieve its objective (reach it's goal)
  - Selecting an action based on goals:
    - From a single action (simple)
    - Combination of several actions required (Search and planning involved)
  - Decision making of this kind is fundamentally different from the condition:
    - Consider both "What will happen if I do such-and-such?" and "Will that make me happy?
    - E.g. Car braking in reflex agents vs goal



Types of agent programs

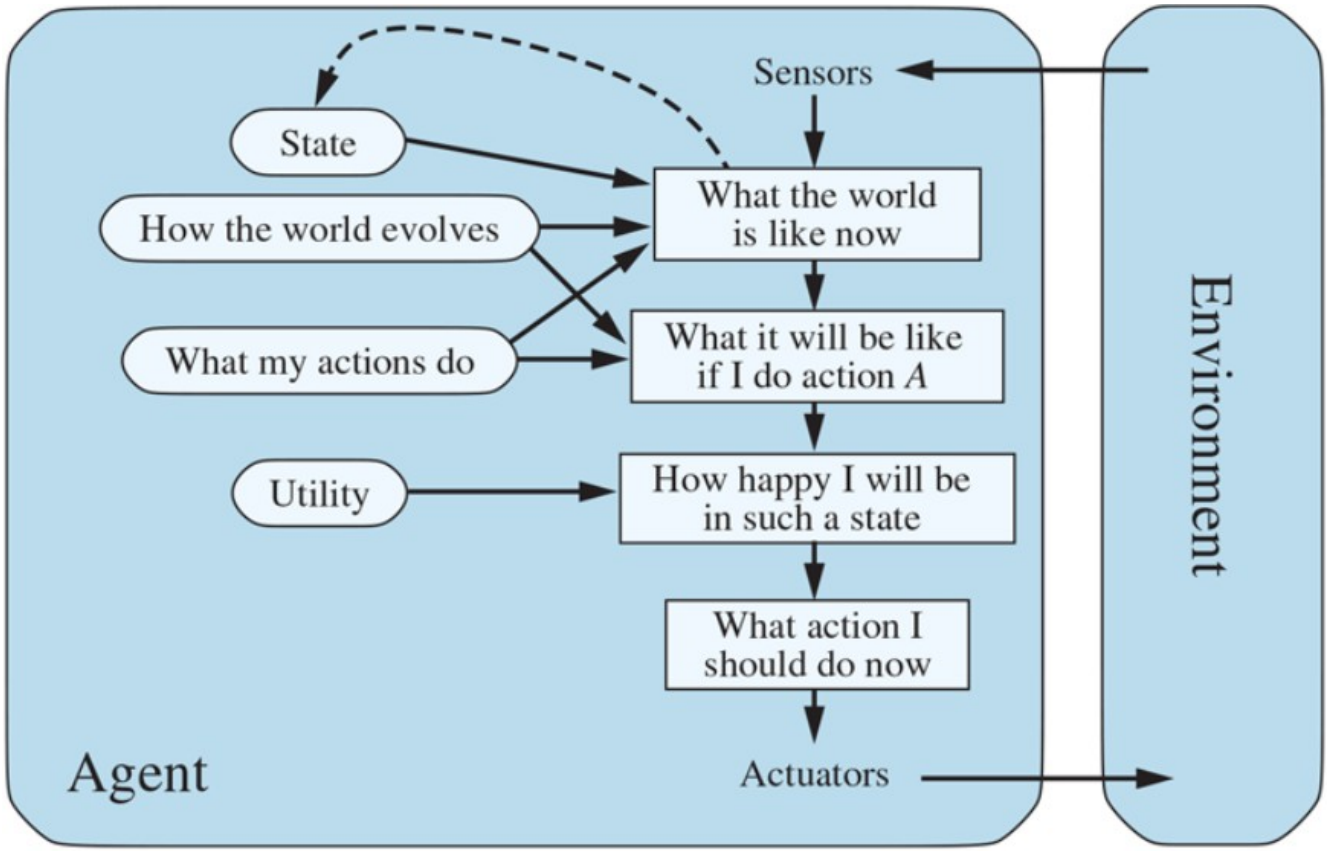
- Based on degree of perceived intelligence (from our perspective) and capability
  - ***Utility-based Agent****: here we're concerned not only about goals but the quality of action taken*
    - A more refined action – performance metrics required (utility function)
    - Use a world model along with utility function that influences its preferences among the states of that world.
    - It chooses the action that leads to the best expected utility
  - Advantages:

    When there are conflicting goals,

    - Only some of the goals but not all can be achieved
    - Utility describes the appropriate trade-off

When there are several goals

- None of them are achieved **<u>certainly</u>**
- Utility provides a way for the decision-making



Types of agent programs

- Based on degree of perceived intelligence (from our perspective) and capability
  - ***Learning Agent**:*
    - is an agent that augments the performance element which determines actions from percept sequences with: (Four conceptual components)
  - **Learning element** : makes improvements to the agent's knowledge
  - **Critic:** gives feedback to the learning element based on an external performance standard (from user or examples, good or not?)
  - **Problem generator:** suggests actions that lead to new and informative experiences.
  - **Performance element:** Selecting external actions or what we took for the whole agent above.

- Learning allows the agent to operate in initially unknown environments and to become more competen than its initial knowledge alone might allow

